Microprocessor Pin Referencing Measurement Technique Writeup

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Target Audience:
Engineers, technicians, embedded developers who are relatively comfortable with using lab instruments like oscilloscopes, signal generators, familiar with voltage division, ohm's law, and microcontroller pin referencing concepts.

Context:
In electronics designs, particularly battery constrained embedded designs, it is important to understand how GPIO of all processors are referenced at all times. Importantly, during steady state, but also during transitory states like during boot, reset, while various peripherals are initialized, terminated, etc. Failure to understand these details can result in undesirable behavior such as pin conflicts, or floating/unreferenced pins. Floating/unreferenced pins do not always cause problems, but are specifically problematic if both GPIO and if the internal digital input circuitry of the pin has not been disabled. My preferred best practice to avoid this in firmware is to assign all unused GPIO to output, push pull, low, non-inverted. Still, whenever possible, I encourage measuring pin referencing on products designed regardless of firmware pin referencing review. This is an excellent opportunity even to establish expected pin referencing behavior, measure, and compare. This is a beneficial analysis technique to perform while optimizing products for minimal power consumption as well as it will often highlight where resistive power consumption is happening.

Scope:
This document describes a novel technique used to check pin referencing of products by applying a 5Mohm source resistance square wave to individual pins to assess how the pins are referenced and in what direction relative to a 5Mohm source resistance.

Core Concept:

An approximate 5Mohm source resistance square wave is created using a signal generator, a 10Mohm series resistor, and a 10Mohm oscilloscope probe. The signal generator is configured for High-Z output impedance. The 10Mohm scope probe provides a load to ground. The series resistor and the scope probe form a voltage divider, reducing the square wave by approximately half. Together, these two resistances create an effective 5Mohm source resistance at the measurement node, which is then used to probe microprocessor pins. The signal generator peak voltage is set to 2x of the microprocessor's rail voltage since it divides by 2. Since the source resistance is quite high, most microcontrollers can handle sourcing/sinking the current while respecting the pins' limits.

When a pin is probed with a 5Mohm source resistance square wave of matching maximum voltage, how the signal reacts indicates how the corresponding micro pin is referenced.

If the pin were output low (shorted to ground relative to 5Mohm), the signal will show 0V DC, with no square wave.

If the pin were driven output high (push-pull), the square wave will turn into a flat voltage at whatever rail voltage that pin is driven to.

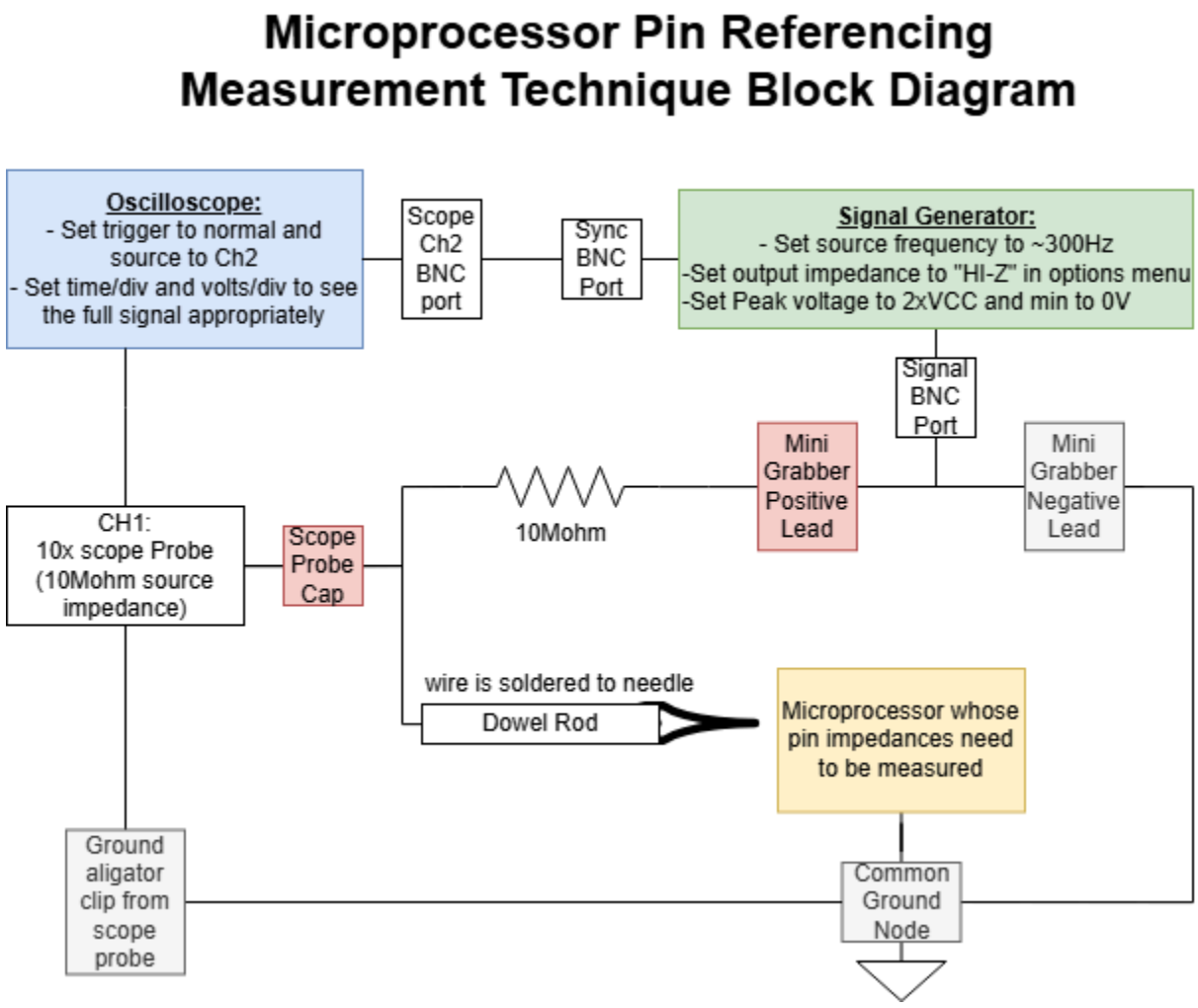
If the pin has a pull-up or pull-down, the signal will be pulled towards that corresponding 0V or rail voltage, but the square wave will be still visible according to the voltage division between 5Mohm and the corresponding pull-up or pull-down.

If the pin is unreferenced and floating, the signal will appear unchanged by the pin being probed.

Equipment needed:

- Oscilloscope with a standard 10x magnification scope probe (most modern scopes support this and come with 10x probes)
- Signal Generator capable with ability to set source impedance to "high-z" (most decent signal generators)
- BNC to BNC cable
- BNC to mini grabber cable
- 10Mohm resistor
- Wooden dowel with a fine pointed needle pressed into the end

Setup:



#TODO: A picture of this instrumented on my home lab bench

#TODO: Scope waveform captures showing what the baseline square wave looks like (high impedance) vs output low (a flat line at 0V), output high (a flat line at VCC), and how pull-ups and pull-downs show the signal divided up or down towards GND or VCC.